

4.13.61

EE25BTECH11057 - Rushil Shanmukha Srinivas

Problem: A rectangle $PQRS$ has its side $\mathbf{PQ}$ parallel to the line $y = mx$ and vertices $\mathbf{P}, \mathbf{Q}$ and $\mathbf{S}$ on the lines $y = a, x = b$ and $x = -b$, respectively. Find the locus of the vertex $\mathbf{R}$.

Solution:

Given, $\mathbf{P}, \mathbf{Q}$ and $\mathbf{S}$ lie on the lines $y = a, x = b$ and $x = -b$.

$$\mathbf{n}_1^\top \mathbf{P} = a, \mathbf{n}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (0.1)$$

$$\mathbf{n}_2^\top \mathbf{Q} = b, \mathbf{n}_2^\top \mathbf{S} = -b, \mathbf{n}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (0.2)$$

$$\mathbf{m} = \begin{pmatrix} 1 \\ m \end{pmatrix}, \mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}, \mathbf{m} \perp \mathbf{n} \quad (0.3)$$

As $\mathbf{PQ}$ is parallel to $\mathbf{m}$ and $\mathbf{PS}$ is parallel to $\mathbf{n}$

$$\mathbf{Q} - \mathbf{P} = \mathbf{R} - \mathbf{S} = k_1 \mathbf{m}, \mathbf{S} - \mathbf{P} = k_2 \mathbf{n} \quad (0.4)$$

$$\mathbf{n}_2^\top \mathbf{m} = 1, \mathbf{n}_2^\top \mathbf{n} = -m \quad (0.5)$$

$$\mathbf{n}_1^\top \mathbf{m} = m, \mathbf{n}_1^\top \mathbf{n} = 1 \quad (0.6)$$

$$\mathbf{Q} = \mathbf{P} + k_1 \mathbf{m} \implies \mathbf{n}_2^\top \mathbf{Q} = \mathbf{n}_2^\top \mathbf{P} + k_1 \mathbf{n}_2^\top \mathbf{m} = b \quad (0.7)$$

$$\mathbf{S} = \mathbf{P} + k_2 \mathbf{n} \implies \mathbf{n}_2^\top \mathbf{S} = \mathbf{n}_2^\top \mathbf{P} + k_2 \mathbf{n}_2^\top \mathbf{n} = -b \quad (0.8)$$

Subtracting (0.7) and (0.8)

$$k_1 + mk_2 = 2b \quad (0.9)$$

$$\mathbf{R} = \mathbf{P} + k_2 \mathbf{n} + k_1 \mathbf{m} \implies \mathbf{n}_1^\top \mathbf{R} = \mathbf{n}_1^\top \mathbf{P} + k_2 \mathbf{n}_1^\top \mathbf{n} + k_1 \mathbf{n}_1^\top \mathbf{m} \quad (0.10)$$

$$\mathbf{n}_1^\top \mathbf{R} = a + k_2 + k_1 m \quad (0.11)$$

$$\mathbf{R} = \mathbf{P} + k_2 \mathbf{n} + k_1 \mathbf{m} \implies \mathbf{n}_2^\top \mathbf{R} = \mathbf{n}_2^\top \mathbf{P} + k_2 \mathbf{n}_2^\top \mathbf{n} + k_1 \mathbf{n}_2^\top \mathbf{m} \quad (0.12)$$

$$\mathbf{n}_2^\top \mathbf{R} = b + k_2 + k_1 m \quad (0.13)$$

$$\mathbf{n}_2^\top \mathbf{R} = b - mk_2 \implies k_2 = \frac{b - \mathbf{n}_2^\top \mathbf{R}}{m} \quad (0.14)$$

$$k_1 + mk_2 = 2b \implies k_1 + b - \mathbf{n}_2^\top \mathbf{R} = 2b \quad (0.15)$$

$$k_1 = b + \mathbf{n}_2^\top \mathbf{R} \quad (0.16)$$

$$\mathbf{n}_1^\top \mathbf{R} = a + k_2 + k_1 m \implies \mathbf{n}_1^\top \mathbf{R} = a + \frac{b - \mathbf{n}_2^\top \mathbf{R}}{m} + mb + m\mathbf{n}_2^\top \mathbf{R} \quad (0.17)$$

$$\mathbf{n}_1^\top \mathbf{R} = a + b \left(m + \frac{1}{m} \right) + \mathbf{n}_2^\top \mathbf{R} \left(m - \frac{1}{m} \right) \quad (0.18)$$

$$\mathbf{n}_2^\top \mathbf{R} \left(\frac{1}{m} - m \right) + \mathbf{n}_1^\top \mathbf{R} = a + b \left(m + \frac{1}{m} \right) \quad (0.19)$$

$$\begin{pmatrix} \frac{1}{m} - m & 1 \end{pmatrix} \begin{pmatrix} \mathbf{n}_2^\top \mathbf{R} \\ \mathbf{n}_1^\top \mathbf{R} \end{pmatrix} = a + b \left(m + \frac{1}{m} \right) \quad (0.20)$$

$$\mathbf{R} = \begin{pmatrix} \mathbf{n}_2^\top \mathbf{R} \\ \mathbf{n}_1^\top \mathbf{R} \end{pmatrix} \quad (0.21)$$

Let

$$c = a + b \left(m + \frac{1}{m} \right) \text{ and } \mathbf{n}_3 = \begin{pmatrix} \frac{1}{m} - m \\ 1 \end{pmatrix} \quad (0.22)$$

The locus of the vertex $\mathbf{R}$ is

$$\mathbf{n}_3^\top \mathbf{R} = c \quad (0.23)$$

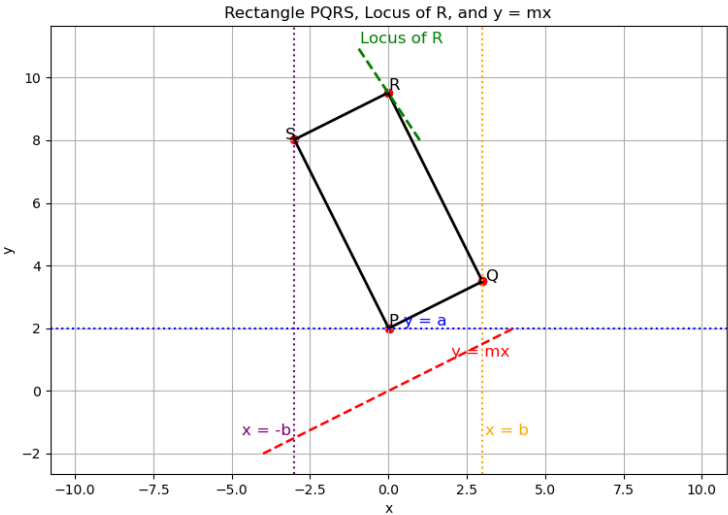


Fig: Representation of Rectangle and Locus